



INNOVATION. AUTOMATION. ANALYTICS

PROJECT ON

Music Store Management System

About Me :

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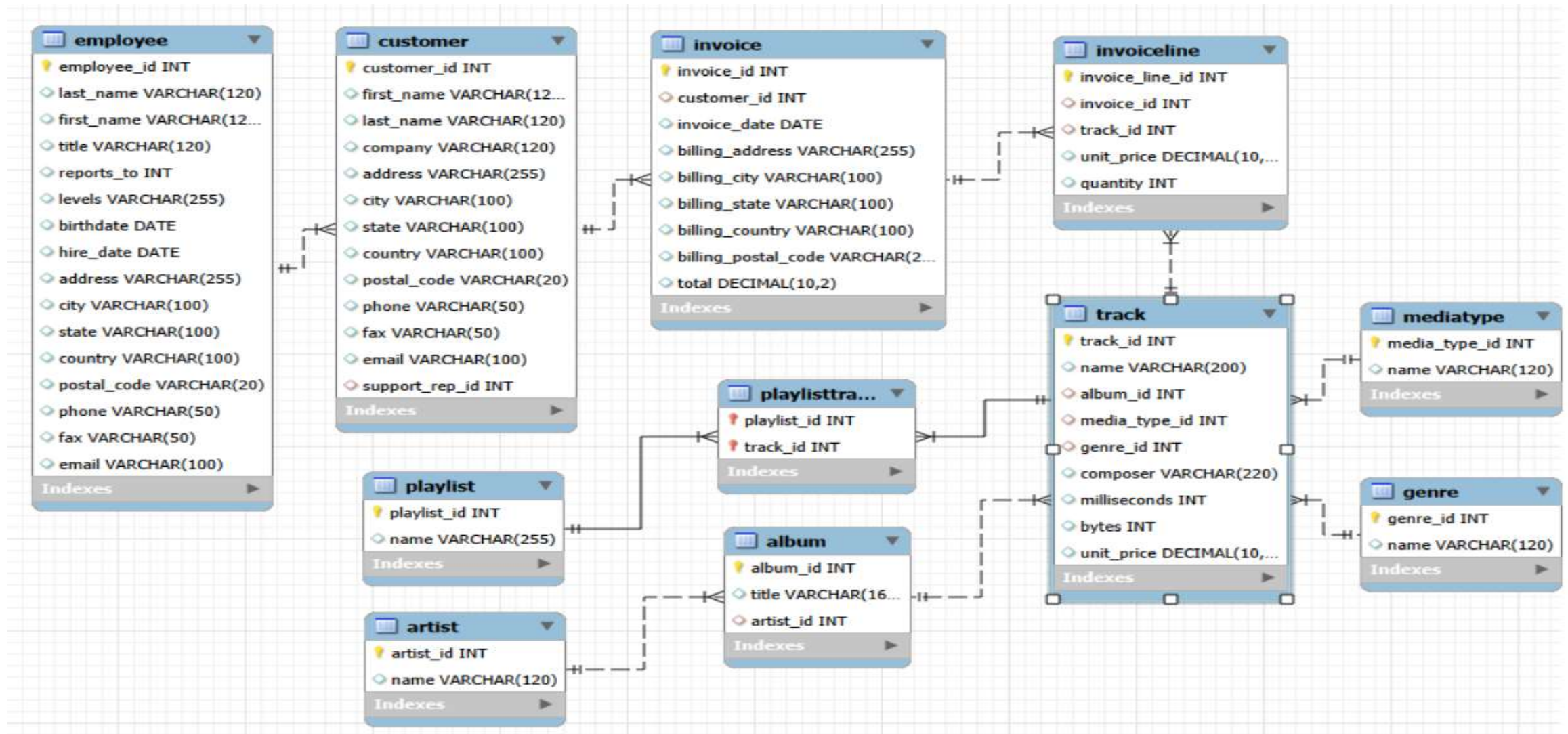
Agenda

- Objective of the Project
- Key analysis questions (use cases)
- Q&A Slide
- Our Experience/Challenges working
- Conclusion

Objective of the Project

- The objective of this project is to **analyze music store data** and gain insights into customer purchasing behavior and sales performance.
- The project aims to identify **high-value customers** and analyze their spending patterns across different countries.
- It also helps evaluate **regional trends** and **customer preferences** to better understand market demand.
- By using SQL queries, the project transforms raw data into meaningful insights for improved business strategies.
- Overall, the goal is to support better decision-making by identifying trends, patterns, and business opportunities within the music store dataset.

Schema Diagram



1. Who is the **senior-most employee** based on **job title**?

Query:

```
select first_name,last_name,title,hire_date
from employee
order by hire_date
limit 1;
```

Output:

Result Grid  Filter Rows: Export: 				
	first_name	last_name	title	hire_date
▶	Nancy	Edwards	Sales Manager	2016-05-01

Key Insight :

- Nancy Edwards, working as a Sales Manager, is the longest-serving employee based on the earliest hire date, indicating significant experience and organizational knowledge.

2. Which countries have the most Invoices?

Query:

```
select billing_country, count(billing_country) as total_bill
from invoice
group by billing_country
order by total_bill desc
limit 5;
```

Output:

Result Grid			Filter Rows:
	billing_country	total_bill	
▶	USA	131	
	Canada	76	
	Brazil	61	
	France	50	
	Germany	41	

Key Insight :

- USA has the highest number of invoices (131), followed by Canada and Brazil, indicating these are the strongest customer markets for the music store.

3. What are the top 3 values of total invoice?

Query:

```
select invoice_id,  
       billing_country,  
       total  
from invoice  
order by total desc  
limit 3;
```

Output:

Result Grid   Filter Rows:			
	invoice_id	billing_country	total
▶	183	France	23.76
	92	Canada	19.80
	31	Canada	19.80
•	NULL	NULL	NULL

Key Insight :

- France recorded the highest invoice value (23.76), while Canada had two high-value invoices (19.80), indicating strong customer spending.

4. Which city has the best customers? - We would like to throw a promotional Music Festival in the city we made the most money. Write a query that returns one city that has the highest sum of invoice totals. Return both the city name & sum of all invoice totals

Query:

```
select billing_city, sum(total) as total_sum
from invoice
group by billing_city
order by total_sum desc limit 1
;
```

Output:

Result Grid			Filter Rows:
	billing_city	total_sum	
▶	Prague	273.24	

Key Insight :

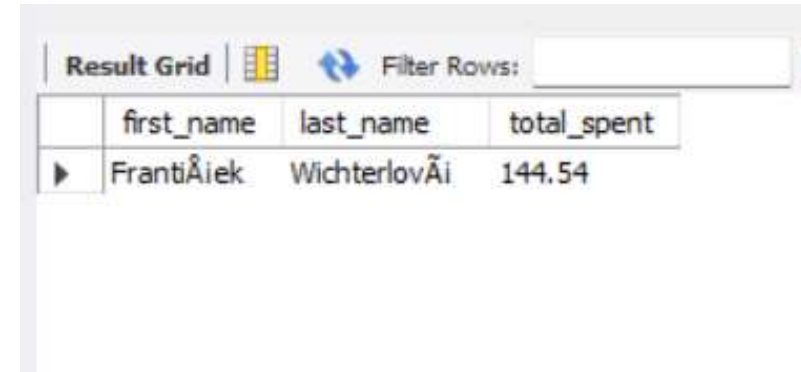
- Prague generated the highest total revenue (273.24), making it the city with the best customers and the ideal location for hosting a promotional Music Festival.

5. Who is the best customer? -The customer who has spent the most money will be declared the best customer. Write a query that returns the person who has spent the most money

Query:

```
select first_name,last_name,sum(total) as total_spent
from customer as c
join invoice as inv
on c.customer_id = inv.customer_id
group by first_name,last_name
order by total_spent desc
limit 1;
```

Output:



The screenshot shows a database interface with a 'Result Grid' tab. It contains a table with three columns: 'first_name', 'last_name', and 'total_spent'. The first row of data shows 'František' as the first name and 'Wichterlová' as the last name, with a 'total_spent' of 144.54. Above the table is a 'Filter Rows:' input field.

	first_name	last_name	total_spent
▶	František	Wichterlová	144.54

Key Insight :

- František Wichterlová is the best customer, with the highest total spending of 144.54, making them the most valuable customer of the music store.

6. Write a query to return the email, first name, last name, & Genre of all Rock Music listeners. Return your list ordered alphabetically by email starting with A

Query:

```
select distinct email, first_name, last_name, g.name
from customer c
join invoice i
on c.customer_id = i.customer_id
join invoiceline inl
on i.invoice_id = inl.invoice_id
join track as t
on t.track_id = inl.track_id
join genre as g
on g.genre_id = t.genre_id
where g.name = "Rock"
order by c.email ;
```

Output:

Result Grid					Filter Rows:	Export:
	email	first_name	last_name	name		
▶	aaronmitchell@yahoo.ca	Aaron	Mitchell	Rock		
	alero@uol.com.br	Alexandre	Rocha	Rock		
	astrid.gruber@apple.at	Astrid	Gruber	Rock		
	bjorn.hansen@yahoo.no	Björn	Hansen	Rock		
	camille.bernard@yahoo.fr	Camille	Bernard	Rock		
	daan.peeters@apple.be	Daan	Peeters	Rock		
	diego.gutierrez@yahoo.ar	Diego	Gutiérrez	Rock		
	dmiller@comcast.com	Dan	Miller	Rock		
	dominiquelefebvre@gmail.com	Dominique	Lefebvre	Rock		
	edfrancis@yahoo.ca	Edward	Francis	Rock		
	eduardo@woodstock.com.br	Eduardo	Martins	Rock		
	ellie.sullivan@shaw.ca	Ellie	Sullivan	Rock		
	emma_jones@hotmail.com	Emma	Jones	Rock		
	enrique_munoz@yahoo.es	Enrique	Muñoz	Rock		
	fernadaramos4@uol.com.br	Fernanda	Ramos	Rock		

Key Insight :

- Customers with emails starting with "A" actively listen to Rock music, representing a targeted segment of Rock listeners.

7. Let's invite the artists who have written the most rock music in our dataset. Write a query that returns the Artist name and total track count of the top 10 rock bands

Query:

```
select art.name, count(track_id) as total_track
from artist as art
join album as al
on art.artist_id = al.artist_id
join track as t
on al.album_id = t.album_id
join genre as g
on g.genre_id = t.genre_id
where g.name = "Rock"
group by art.name
order by total_track desc
limit 10 ;
```

Output:

Result Grid			Filter Rows:
	name	total_track	
▶	AC/DC	18	
	Aerosmith	15	
	Audioslave	14	
	Led Zeppelin	14	
	Alanis Morissette	13	
	Alice In Chains	12	
	Frank Zappa & Captain Beefheart	9	
	Accept	4	

Key Insight :

- AC/DC has the highest number of Rock tracks (18), making it the leading Rock artist in the dataset and a strong choice for Rock music promotions or event invitations.

8. Return all the track names that have a song length longer than the average song length.- Return the Name and Milliseconds for each track. Order by the song length, with the longest songs listed first

Query:

```
select name,milliseconds
from track
where milliseconds > (
    select avg(milliseconds)
    from track
)
order by milliseconds desc
;
```

Output:

	name	milliseconds
▶	How Many More Times	711836
	Advance Romance	677694
	Sleeping Village	644571
	You Shook Me(2)	619467
	Talkin' 'Bout Women Obviously	589531
	Stratus	582086
	No More Tears	555075
	The Alchemist	509413
	Wheels Of Confusion / The Straightener	494524
	Book Of Thel	494393
	You Oughta Know (Alternate)	491885
	Terra	482429
	Snoopy's search-Red baron	456071
	Sozinho (Hitmakers Classic Mix)	436636
	Master Of Puppets	436453
	Stone Crazy	433397
	Snowblind	420022

Key Insight :

- Several tracks exceed the average song length, with “How Many More Times” being the longest track in the dataset .

9. Find how much amount is spent by each customer on artists? Write a query to return customer name, artist name and total spent

Query:

```
select c.first_name , c.last_name , a.name,  
       sum(ino.unit_price * ino.quantity) as total_spent  
from customer c join invoice i on c.customer_id=i.customer_id  
join invoiceline ino on i.invoice_id=ino.invoice_id  
join track t on ino.track_id=t.track_id  
join album al on al.album_id = t.album_id  
join artist a on al.artist_id=a.artist_id  
group by c.customer_id, c.first_name,  
         c.last_name, a.artist_id, a.name  
order by total_spent desc ;
```

Output:

	first_name	last_name	name	total_spent
▶	Steve	Murray	AC/DC	17.82
	Jennifer	Peterson	Aerosmith	14.85
	Mark	Taylor	Aerosmith	14.85
	Fernanda	Ramos	Antônio Carlos Jobim	13.86
	Leonie	Köhler	Audioslave	13.86
	Edward	Francis	Alanis Morissette	12.87
	Emma	Jones	Alanis Morissette	12.87
	João	Fernandes	Alanis Morissette	12.87
	Victor	Stevens	Alice In Chains	11.88
	Kathy	Chase	AC/DC	10.89
	Phil	Hughes	AC/DC	10.89
	Stanisław	Wójcik	Buddy Guy	10.89
	Stanisław	Wójcik	AC/DC	9.90
	François	Tremblay	Apocalyptica	7.92
	Frank	Ralston	AC/DC	7.92
	Kathy	Chase	Black Sabbath	7.92
	Luís	Gonçalves	AC/DC	7.92
	Frank	Ralston	Black Sabbath	6.93

Key Insight :

- Customers show different spending patterns across artists, with Steve Murray spending the highest amount on AC/DC (17.82), indicating strong artist preference.

10. We want to find out the most popular music Genre for each country. We determine the most popular genre as the genre with the highest amount of purchases. Write a query that returns each country along with the top Genre. For countries where the maximum number of purchases is shared, return all Genres

Query:

```
WITH genre_purchases AS (  
    SELECT  
        c.country, g.name AS genre,  
        SUM(il.quantity) AS total_purchases,  
        RANK() OVER (  
            PARTITION BY c.country  
            ORDER BY SUM(il.quantity) DESC  
        ) AS rank_genre  
    FROM customer as c  
    JOIN invoice as i  
        ON c.customer_id = i.customer_id  
    JOIN invoiceline as il  
        ON i.invoice_id = il.invoice_id  
    JOIN track as t  
        ON il.track_id = t.track_id  
    JOIN genre as g  
        ON t.genre_id = g.genre_id  
    GROUP BY c.country, g.name  
)  
SELECT  
    country, genre, total_purchases  
FROM genre_purchases  
WHERE rank_genre = 1  
ORDER BY country ;
```

Output:

	country	genre_name	total_purchases
▶	Argentina	Rock	1
	Australia	Rock	18
	Austria	Rock	6
	Belgium	Rock	5
	Brazil	Rock	26
	Canada	Rock	57
	Chile	Rock	7
	Czech Republic	Rock	14
	Denmark	Rock	6
	Finland	Rock	6
	France	Rock	26
	Germany	Rock	28
	Hungary	Rock	4
	India	Rock	13
	Ireland	Rock	2
	Italy	Rock	3
	Netherlands	Rock	6
	Norway	Metal	2
	Poland	Rock	14
	Portugal	Rock	23
	Spain	Metal	4

Key Insight :

- Rock is the most popular genre in the majority of countries, showing its global dominance and widespread customer preference.
- A few countries like Norway and Spain prefer Metal, indicating regional variation in music taste.
- Countries such as Canada, Brazil, and Germany have higher purchase counts, highlighting strong demand and engagement with top genres in these regions.

11. Write a query that determines the customer that has spent the most on music for each country. Write a query that returns the country along with the top customer and how much they spent. For countries where the top amount spent is shared, provide all customers who spent this amount

Query:

```
SELECT country, customer_name, total_spent
FROM (
    SELECT
        c.country,
        c.first_name ,c.last_name AS customer_name,
        SUM(i.total) AS total_spent,
        RANK() OVER (
            PARTITION BY c.country
            ORDER BY SUM(i.total) DESC
        ) AS rnk
    FROM customer c
    JOIN invoice i
        ON c.customer_id = i.customer_id
    GROUP BY c.country, c.customer_id
) ranked_data
WHERE rnk = 1
ORDER BY country;
```

Output:

	country	customer_name	total_spent
▶	Argentina	Gutiérrez	39.60
	Australia	Taylor	81.18
	Austria	Gruber	69.30
	Belgium	Peeters	60.39
	Brazil	Gonçalves	108.90
	Canada	Tremblay	99.99
	Chile	Rojas	97.02
	Czech Republic	Wichterlová	144.54
	Denmark	Nielsen	37.62
	Finland	Hämäläinen	79.20
	France	Girard	99.99
	Germany	Zimmermann	94.05
	Hungary	Kovács	78.21
	India	Pareek	111.87
	Ireland	O'Reilly	114.84
	Italy	Mancini	50.49

Key Insight :

- Each country has a top-spending customer, with the Czech Republic (144.54) and Ireland (114.84) showing some of the highest customer spending.
- Customer spending varies across countries, highlighting differences in purchasing behavior and customer value.
- Identifying top customers helps businesses focus on high-value customer retention and targeted marketing.

Our Experience / Challenges Working

- There were many tables in the database, so connecting them using JOIN was sometimes challenging.
- Understanding table relationships and selecting the correct columns for queries took time.
- Using SQL concepts like CTE and Window Functions RANK() was difficult at first, but helped us solve complex problems.
- Writing long queries and fixing errors improved our SQL and problem-solving skills.
- Overall, this project improved our practical understanding of SQL and data analysis.

Conclusion

- The analysis shows that sales are mainly concentrated in a few important countries and cities.
- Rock is the most popular music genre, with some differences in regional preferences.
- A small group of customers contributes significantly to total sales.
- Customer buying behavior is influenced by their favorite artists and genres.
- SQL analysis helps convert raw data into useful insights for better business decisions and growth.

THANK
YOU

